

Computational methods in Astrophysics

Homework #4

Programs can be written in c, c++, Matlab, Python or Python notebook (preferably the last). In addition to the program, you should have a writeup that contains plots requested, answers to any analytical or explanation questions, and a short description of your code and how to run it. This can be done in, e.g., LATEX, WORD, etc. If your program requires a compilation to run, add the executable file as well. Code and writeup should be submitted to the Moodle site.

- 1) *Relaxation methods for the Poisson equation.*
 - a. Write a program that solves the two-dimensional Poisson equation in a square geometry with the Dirichlet boundary conditions $\Phi = 0$ at the boundaries using one of the relaxation methods discussed in class. Map the potential for a single charge at the center of the system. Compare with the potential for a charge in free space (recall that in two dimensions this charge is a line charge and not a point charge).
 - b. Modify your program to use periodic boundary conditions. Compare the result to part (a)
- 2) *Solutions to the Burger equation:* consider two different first-order finite-difference methods for Burgers' equation:

$$u_t + uu_x = 0$$

As we saw in class, this is a nonlinear equation which can be written in a conservative form as:

$$u_t + \left[\frac{1}{2} u^2 \right]_x = 0$$

Although these forms are analytically equivalent, the numerical solutions differ. For the following, use outflow boundary conditions.

- a. Write a first order, upwind differentiate scheme for the first form of Burgers' equation. Write a program to solve the equation with initial conditions:

$$u(x, t = 0) = \begin{cases} 2 & \text{if } x < 0.5 \\ 1 & \text{if } x \geq 0.5 \end{cases}$$

These are the same initial conditions used in the shock demonstration we saw in class. Plot the shock at different times, similar to how we did it in class. Make sure you use adequate CFL conditions,

- b. Do the same for the conservative form: write a first order upwind differentiate scheme, solve it numerically with the same initial conditions as in (a) and plot the shock evolution.
- c. Measure the speed of the shock in both methods in three different resolutions (e.g. by finding the first zone where the velocity drops below some value). How does the shock speed differ with resolution? Which method gets the correct shock speed? Why?

- 3) *Motion of a charged particle in an electromagnetic field*: Implement the leapfrog algorithm to numerically solve the 2D ODE equation of motion of a charged particle in an EM field:

$$\begin{aligned}\frac{d\vec{r}(x,y)}{dt} &= \vec{v}(x,y) \\ \frac{d\vec{v}(x,y)}{dt} &= \left(\vec{E}(\vec{r},t) + \vec{v} \times \vec{B}(\vec{r},t) \right),\end{aligned}$$

where $\vec{E}(\vec{r},t)$ and $\vec{B}(\vec{r},t)$ are known (here we assume that $\frac{q}{m} = 1$). Use Boris' algorithm to advance $\vec{v}(x,y)$ in time.

- a. Apply the program for a case of a uniform magnetic field, $\vec{B} = 1\hat{z}$ and no electric field. Use an initial condition $\vec{x}_0 = (0,1)$ and $\vec{v}_0 = (1,0)$. Plot the path of the particle. Calculate the radius and frequency of the motion and compare them with the theoretical Larmor radius and frequency.
- b. Apply the program for a case of a uniform magnetic field, $\vec{B} = 1\hat{z}$ and a uniform electric field $\vec{E} = -0.5\hat{y}$ with the same initial conditions as in (a). Plot the path of the particle. Calculate the drift velocity and compare it with the theoretical expectation.